



PAVO

Offline Learning Companion

A gamified-but-academic learning app for students, teachers, and parents that works fully offline. Lessons, quizzes, spaced-repetition study, and class insight - all on-device, synced between phones by QR codes and device-to-device transfer.

- **Works fully offline**
- **No account needed to learn**
- **Built for the classroom**
- **Private by default - data stays on device**

1. What Pavo is

Pavo is an offline-first mobile learning application (React Native + Expo) built for low-connectivity classrooms. Every core activity - reading lessons, taking quizzes, practising flashcards, and tracking progress - runs entirely on the device with no internet required. When connectivity exists it can enhance the experience, but it is never a prerequisite.

The product serves three roles from one app: students learn and self-assess; teachers build a class picture by scanning student QR reports and author or hand-deliver lessons; parents receive a plain-language weekly digest. A friendly pixel peacock, Pavo, acts as both guide and a five-phase growth indicator that matures as the learner progresses.

Who it is for

- Students: Grade-5 learners on phones, often sharing devices and working without a reliable connection.
- Teachers: Classroom teachers who need a live class picture, offline, assembled from what students bring to them.
- Parents and guardians: Caregivers who want a short, credible weekly summary of how their child is doing.

2. Design language - the Peacock system

Pavo's identity is drawn from peacock plumage: a deep-teal primary, an ocean-blue secondary, and a sparing peacock-eye gold accent reserved for earned moments. The tone is a high-end academic companion - structured and calm, credible to a teacher, with warmth and earned progression that keeps a student returning. It is deliberately not a toy and not an arcade.

- Single source of truth: All color, type, spacing, radius, elevation, and motion live in one design-token file; screens never hard-code them.
- Typography: One optically-tracked type scale on the system font, so large headings read tight and editorial with no custom font shipped.
- Components: A shared kit - cards, hero surfaces, gradients, stat and action tiles, progress bars and rings, segmented controls, callouts, list rows - gives every screen the same rhythm.
- Mascot: A crisp pixel-art peacock that grows through five phases (Hatchling, Fledgling, Plumed, Brilliant, Radiant) from real progress.
- Accessibility: WCAG 2.1 AA text pairings, 44pt+ touch targets, roles/labels/states on interactive elements, and full reduced-motion support on every animation.

3. Student features

Onboarding and identity

- Profile creation and PIN sign-in: Students create a local profile and sign in with a 4-digit PIN; everything stays on the device.
- Guardian-assisted setup: An optional guardian acknowledgement step during first-time setup.
- Learning-style assessment: A short guided assessment derives a primary learning style and a recommended default format.
- Adaptive format: Pavo recommends text, audio, visual, or kinesthetic based on recent completed work, with

a manual override at any time.

Home, progression, and gamification

- Learner hero: The home anchor shows the growth peacock inside a gold level ring, the learner's scholarly rank and phase, and level progress.
- Study points and levels: Points are earned transparently from completed modules, quiz attempts, and mastery; levels rise on a gentle curve through ranks Novice, Apprentice, Scholar, Adept, and Luminary.
- Milestones: Eight achievement badges (first lesson, five deep, halfway, full sweep, sharp, mastery, persistent, cards clear). Locked badges stay visible as an invitation and never light up unearned.
- Dashboard: Completion, average score, cards due, and quiz attempts as stat tiles, plus deadlines, today's focus (strong area and practice next), and quick actions.
- Note: Every number is derived at render time from real activity - nothing is invented or persisted as fake state.

Modules and lessons

- Offline library: Grade-5 Science, Math, and English content bundled for offline use, with subject filtering and a teacher-added category.
- Markdown reader: A native Markdown lesson reader with local WebP images and comfortable reading typography.
- Read aloud: On-device text-to-speech reads a lesson, with an optional cloud-generated voice when online.
- Inline recall: Tap a highlighted term or select any phrase to turn it into an instant fill-in-the-blank retrieval check.

Quizzes and results

- Unlimited attempts: Learners can retake quizzes freely; each attempt records the chosen learning format.
- Local timing: Per-question and total timing are captured on-device.
- Results: A score with mastery level, strong and weak topics, a question-by-question breakdown, and correct answers for missed items.
- Celebration: A tasteful, skippable confetti moment for strong scores, suppressed under reduced motion.

Study tools

- Spaced repetition: SM-2 scheduling with 0-5 ratings brings flashcards back exactly when they are most useful.
- Active-recall flashcards: Answers stay hidden until the learner commits to a recall.
- Retrieval-practice quizzes: Feedback is delayed until completion.
- Interleaving, self-explanation, blurring: Additional evidence-based techniques with clear 'why it works' and 'how to use it here' guidance.
- Pomodoro focus timer: Configurable work/break lengths with a queue sized from the learner's historical response times; its review events feed the same SM-2 schedule.
- Custom review sets: Student- or teacher-created sets from existing or new prompt-and-answer items, private or shared with the class.

Sharing and reports

- Profile QR: A scannable profile QR the learner shows once to join a teacher's active section.
- Quiz-report QR: Offline quiz reports are encoded as QR codes, automatically split into multiple parts when dense, carrying timing and missed-answer detail only - never question text or correct answers.
- Reports screen: A history of attempts, each reopenable as its QR.
- Parent digest: A weekly summary of completed modules, study-time trend, current format preference, and a home suggestion.

4. Teacher features

Class dashboard and record book

- Local class dashboard: Built entirely from scanned student reports - class average (latest attempt per module), a teacher-only leaderboard, and commonly-missed concepts.
- Support threshold: A configurable struggling threshold (50/60/70%) flags learners by low score or a three-attempt declining trend.
- Record book: A searchable roster with each learner's average, attempts, completed modules, practice priority, and recommended format with a confidence percentage.
- Learner detail: A full profile with stat tiles, per-module breakdown, strong and weak areas, and quiz history.

Building and delivering content

- Scanner: Camera-based QR scanning that assembles multipart reports with clear idle, scanning, success, and error states.
- Sections: Create and activate class sections; build a roster by scanning student profile QR codes.
- Assignment builder: Compose module or quiz tasks with due dates and hand them out as an assignment QR.
- Module author: A Markdown authoring tool with a live readiness summary that packages a lesson for device-to-device transfer.
- Nearby transfer: Send modules to a nearby device over Bluetooth or local Wi-Fi with a pairing code, SHA-256 verification, progress, and resume.
- Gurobot (planned): A privacy-preserving teaching assistant for lesson drafts and class summaries; authoring by hand is available today.
- Teacher profile: Faculty details, an active-section snapshot, device mode, teaching-tool shortcuts, and sign out.

5. Offline, sync, and data

- On-device storage: Offline-first SQLite persistence with additive migrations that preserve existing student data; module and transfer files are app-private.
- QR as the sync fabric: Students carry results to teachers as QR codes; Zod validation runs before generation and after scanning, and corrupted or incomplete multipart scans are rejected.
- Device-to-device transfer: Google Nearby Connections on Android and iOS move .wais-module archives (Markdown + WebP + manifest) with integrity checks; imported review items immediately join the scheduler.
- Privacy by design: Raw answers stay local; any AI request excludes names, IDs, birthdays, sections, and raw answers; QR reports omit question text and correct-answer detail.
- Two modes: A lightweight offline mode and a full mode that adds online enhancements - core learning works the same in both.

6. Technology

Framework React Native 0.86 on Expo SDK 57 (prebuild), TypeScript.

Navigation React Navigation 7 - native stack plus bottom tabs.

State Zustand store for session and mode.

Storage expo-sqlite, offline-first with additive migrations.

Graphics react-native-svg for gradients, rings, QR, and the pixel mascot; no image assets required for the UI system.

Native Android and iOS Google Nearby Connections modules for transfer; expo-camera for scanning; expo-speech for read-aloud.

Quality Typechecked with tsc; deterministic review, dashboard, digest, and progression logic covered by unit tests.

7. Trying the demo

From the landing screen, 'Explore a demo classroom' seeds a believable Grade-5 class (Section Mabini) with a teacher, eight learners spanning top performers to those needing support, four modules with quizzes and flashcards, multi-attempt histories, due flashcards, a retrieval-practice bank, upcoming deadlines, and format history - then signs you in as the demo student.

Demo student Number 2026-001, PIN 1234 (Maria Santos).

Demo teacher Faculty T-2026 (Ms. Reyes); open 'I'm a teacher' to view the class dashboard, record book, and leaderboard.

Demonstrating that the QR actually works

The QR codes are real, standards-compliant QR - they encode live app data and decode on a second device. Because a phone cannot scan its own screen, an end-to-end QR demo uses two devices:

- 1. Roster join: On device A (student), open Profile and show 'My profile QR'. On device B (teacher), open Scan and point the camera at it - the learner joins the active section.
- 2. Report submission: On device A, finish a quiz and open 'Show report QR' (scan every numbered part if it splits). On device B, scan them - the teacher's dashboard and record book update from the result.

Single-device shortcut: the seeded demo classroom already contains the scanned results, so the teacher dashboard, leaderboard, and record book are fully populated without any scanning - ideal for showing the teacher experience on one phone. To prove the scan path itself, use two devices (or two simulators) as above, or display a student QR on one screen and scan it with another phone.

8. Feature reference

Student surface

Home Hero with level ring and rank, quick actions, deadlines, today's focus, progress tiles, milestones.

Modules Filterable offline library; subject-spined lesson cards.

Lesson reader Markdown + images, read-aloud, inline recall.

Quiz Format choice, per-question timing, lettered options, step progress.

Results Score, mastery, strong/weak topics, question breakdown, QR.

Flashcards SM-2 review with reveal and 0-5 rating.

Review hub Spaced repetition, retrieval practice, interleaving, custom sets, Pomodoro focus timer.

Reports Attempt history, each reopenable as an offline QR.

Profile Profile QR, default format, device mode, parent digest, receive a module, sign out.

Teacher surface

Overview Class dashboard - average, leaderboard, missed concepts, support threshold, quick actions.

Record book Searchable roster with mastery, status, and format.

Learner detail Per-learner profile, module breakdown, quiz history.

Scan Camera QR scanner with multipart assembly and clear states.

Sections Create/activate sections; build rosters by QR.

Assignment builder Module/quiz tasks with due dates as an assignment QR.

Module author Markdown authoring with a readiness summary and packaging.

Nearby transfer Verified device-to-device module delivery.

Assist (Gurobot) Planned AI helper; hand-authoring available now.

Profile Faculty details, active section, device mode, sign out.

Pavo - learning that stays with you, online or offline. Every figure in the app is calculated on the device; nothing about a learner leaves it without consent.